

Data Representation

Chapter 2: End-to-End Machine Learning Project
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Gurelien Geron, 3e, O'Reilly Media, 2022.

Representing Dataset in Supervised Learning

- In supervised learning, each training example consists of an input-output pair $(x^{(i)}, y^{(i)})$, called the i^{th} training example, where:
 - $x^{(i)}$ is the **input** or **feature vector**, containing multiple features:
$$x^{(i)} = (x_1^{(i)}, x_2^{(i)}, x_3^{(i)}, \dots, x_m^{(i)})$$
Here, m is the number of features, and $x_j^{(i)}$ denotes the j^{th} of the i^{th} example.
 - $y^{(i)}$ is the **output**, called the **target** in regression and **label/class** in classification.
- The size of the training set is n , meaning there are n such examples:
$$(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, \dots, (x^{(n)}, y^{(n)})$$
- The objective of learning is to find a hypothesis function $h: X \rightarrow Y$ that maps input x to a predicted output $h(x)$ or $\hat{y}$, which should be as close as possible to the true value y .

Visualizing Machine Learning Data

1. Feature-target (Input-Output) Plots (Target Space View)

- This type of graph shows the relationship between input features and the output (label/target).
- Helps understand how features influence the output, detect patterns, trends, and noise.
- Such graphs are $m+1$ dimensional, where m is the number of features.

2. Feature-Feature Plots (Feature Space View)

- This type of graph plots features against each other, independent of the output (or sometimes color-coded by output).
- Helps understand the structure of the data, correlations, clusters, separation, and redundancy.
- Features are numeric representation of raw data.
- Feature space is an m -dimensional space where features live (m is the number of features).
- Each data point / training example is an m -dimensional vector in feature space.
- Scatter plots can be used to visualize feature space of at most 3 dimensions.

